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Business Process Procedure Overview
Material reclassification and safety stock calculation (IAM) AA-IN

Definition of terms

IAM	Independent After Market
ADC	Area Distribution Center
GtS	Go-To-Source: Leading dimension Marketing
GtM	Go-to-Market: Leading dimension Sales
CBF	Core Business Field
PDT	Planned Delivery Time
GRT	Goods Receiving Time
PO	Purchasing Order
MRP	Material Requirement Planning
SFA	Sales Forecast Accuracy
CSL	Customer Service Level
LT	Lead Time
TCT	Total Coverage Time
MTO	Make To Order
MTS	Make To Stock

1. Material Classification:

Material Classification defines the assignment of materials to specific planning logic in Bosch IAM, determining the applicable MRP type, stocking strategy, and safety stock parameters that drive inventory and replenishment control.

2. MRP Types:

MRP type defines the planning logic for a material in SAP and controls which planning parameters can be maintained in the material master (e.g. forecast relevance, safety stock, replenishment triggers).

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2.1 Standard MRP Types used:

MRP Type	Planning Logic	Application
VV	Make-to-Stock, statistical forecast-based	Fast-moving parts stocked at all LDCs
SP	Make-to-Stock, manual forecast	Slow-moving parts with demand planning alignment (OFTM & SMS-XX), stocked at BAN and NCR
OP	Make-to-Stock, fixed safety stock	No forecast, PR triggered when stock falls below safety stock (ADS / TCT / VOR)
PD	Make-to-Order	Order-based planning as per globally defined ABC logic
ND	No planning	Obsolete or non-planning-relevant materials

2.2 System Usage Clarification

MRP types directly impact forecasting, replenishment, safety stock calculation, and material flow in the distribution network.

SAP and addONE provide multiple additional MRP Types (e.g. AB, FC, FM, M, P*, R*, V*, Y*, Z*), these are system available only and not used for IAM planning unless explicitly approved.

2.3 Stocking Strategy:

Stocking strategy defines how inventory is positioned in the delivery network to achieve the targeted customer service level while balancing inventory levels and availability at the order entry point.

3. Stocking Policy:

Stocking policy defines the stock level to be maintained at each delivery node based on the assigned MRP type

(Refer Section 2 – MRP Types).

It provides visibility of available and sold inventory and enables proactive inventory management to meet customer demand with shorter lead times, higher efficiency, and optimized inventory cost.

4. Storage and distribution strategy:

Storage and distribution strategy in AA-IN is determined by the **stocking policy** and the assigned **MRP type**

(Refer Section 3 – Stocking Policy and Section 2 – MRP Types).

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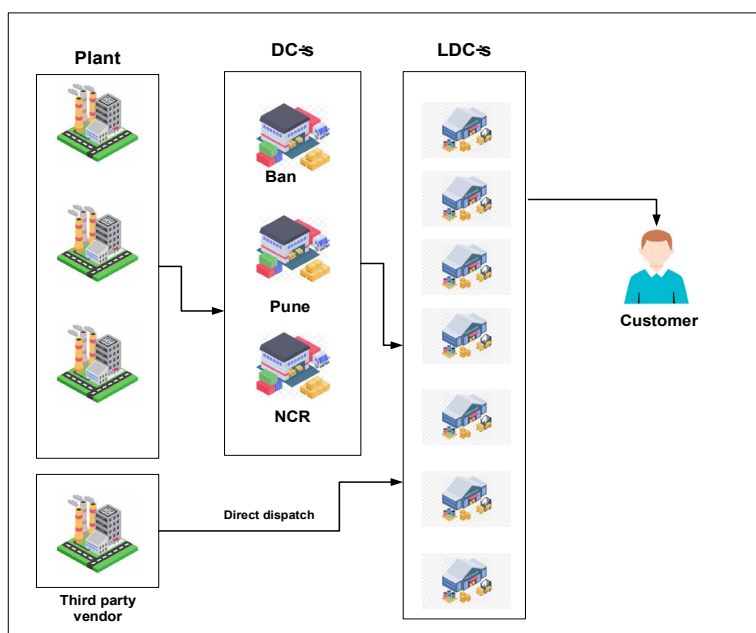
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Based on the targeted service level, materials are classified as **fast movers** or **slow movers**, and material flow is defined accordingly.

For Fast movers:

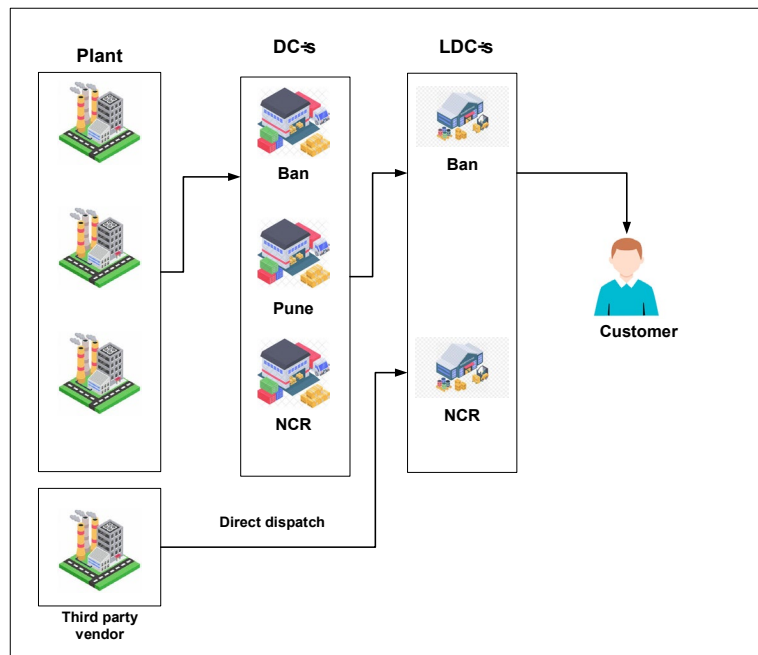
- Applicable for materials with **MRP type VV**
- Stocked across **all LDCs**
- Customer orders are fulfilled from the **nearest LDC** to ensure high availability and shorter lead time



For slow movers:

- Applicable for materials with **MRP types SP, OP, PD, and Imports**
- Stocked only at **designated locations (BAN, NCR)**
- Customer orders are routed to the assigned stocking location based on distribution logic

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4.1 System checks for post MRP changes

Link all the parameters to ERP (SAP) based on standard procedure to update the parameters.

4.1.1 Mapping for order flow -

T-code /RBI2/YV_PLTDET

MRP Type on Plant level										
SOrg.	DChl	Dv	Typ	Plnt	Fixed Plant I...	Plant 1	Plant 2	Plant 3	Plant 4	Plan...
☐ IN20	☐	40	w047	☐	w069	2.				
☐ IN20	G1	40	w049	☐	w069					
☐ IN20	G1	40	w057	☐	w290					
☐ IN20	G1	40	w058	☐	w069					
☐ IN20	G1	40	w059	☐	w069					
☐ IN20	G1	40	w061	☐	w290					
☐ IN20	G1	40	w063	☐	w290					
☐ IN20	G1	40	w067	☐	w290					
☐ IN20	G1	40	VV	☒		1.				
☐ IN2A	G1	40	w212	☐	w227					
☐ IN2A	G1	40	w217	☐	w291					
☐ IN2A	G1	40	w222	☐	w291					
☐ IN2A	G1	40	w223	☐	w227					

1. For VV parts, safety stock is maintained at all locations to fulfill against the customer order from nearest WH.
2. For SP part, safety stock is maintained at two locations only (BAN, NCR). Order placed by customer for SP parts will be transferred to either of two locations based on the shortest distance to customer location and orders are fulfilled.

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4.1.2 Customer mapping for slow/fast moving parts –

Alternative slow mover plant						
SOrg.	DChl	Dv	Customer	Funct	Plnt	Comment
☐ IN20		40	17002706	SP	W058	ST_MAHARASHTRA
☐ IN20	G1	40	17003532	SP	W058	ST_MAHARASHTRA
☐ IN20	G1	40	17003537	SP	W058	ST_MAHARASHTRA
☐ IN20	G1	40	17003538	SP	W058	ST_MAHARASHTRA
☐ IN20	G1	40	17003539	SP	W058	ST_MAHARASHTRA
☐ IN20	G1	40	17003540	SP	W058	ST_MAHARASHTRA
☐ IN20	G1	40	17003541	SP	W058	ST_MAHARASHTRA
☐ IN20	G1	40	17003542	SP	W058	ST_MAHARASHTRA
☐ IN20	G1	40	17003543	SP	W058	ST_MAHARASHTRA
☐ IN20	G1	40	17003544	SP	W058	ST_MAHARASHTRA
☐ IN20	G1	40	17003545	SP	W058	ST_MAHARASHTRA
☐ IN20	G1	40	17003546	SP	W058	ST_MAHARASHTRA
☐ IN20	G1	40	17003547	SP	W058	ST_MAHARASHTRA

Based on distance for few customers exceptions are made in SP parts distribution.

5. Safety stock:

A safety stock is a stock that is used to buffer uncertainties regarding sales development, delivery quantities and delivery dates so that a defined service level can be reached.

Reason for safety stock

- Act as a replenishment trigger
- Used to cater to fluctuation in demand within replenishment lead time

Safety stock is derived based on following

Input	Process step	Output
• Stocking policy (VV, SP) • Demand history (12M) • CSL trend (12M) + Target CSL • TCT, inventory • Customer lead time, source • Distribution network • SFA	Determine optimum safety stock considering the various influencing parameters like demand variations, CSL, Leadtime and forecast accuracy.	Safety stock at warehouse level for relevant MTS parts to ensure on-shelf availability against customer orders.

5.1 Methods for safety stock calculation:

The quantity of safety stock to be stocked has to be defined. To determine the quantity there are different methods for calculation. The different methods have listed below,

Method 1: Based on demand variability without lead time fluctuation.

Demand fluctuates from period to period. In this method we use average demand for specific period and take standard deviation of demand multiply with Z-score without considering lead time.

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Safety stock = Z-score x Standard deviation of demand

Method 2: Based on demand variability with lead time.
In this method demand variation with constant lead time.

Safety stock = Z-score x $\sqrt{\text{Leadtime} \times \text{standard deviation of demand}}$

Method 3: Based on Independent variation in both lead-time and demand.
The demand and lead time both fluctuate from period to period, but the fluctuation in one is completely independent from another.

Safety stock =
Z-score x $\sqrt{(\text{LT} \times (\text{standard deviation demand})^2) + ((\text{standard deviation LT})^2 \times \text{average demand})}$

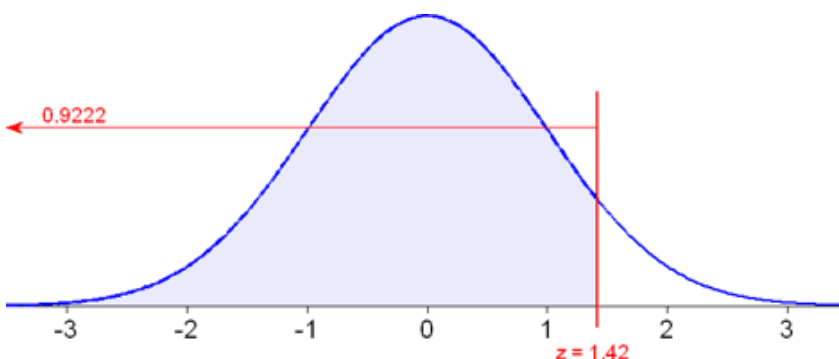
Method 4: Based on dependent variation in LT and demand.
The demand and lead time fluctuate for because of same cause, then variation in one is dependent on other.

Safety stock = (Z-score x $\sqrt{(\text{LT} \times \text{standard deviation of demand})}$) + (Z-score x average demand x standard deviation of LT)

Based on product distribution and use case, method 2 was used for optimal calculation for region - IN (IAM).

5.2 Z-score:

Z-score describes the position of raw score in terms of its distance from the mean, when measured in standard deviation units. The z-score is a single value. It is represented by a vertical line on the graph. That vertical line separates the shaded region from the unshaded region.



Service Level	Z-Score
50	0
85	1.04
90	1.28
95	1.65
98	2.05
99.9	3.09

Pre-defined table for Z-score and desired service level.

Current CSL	Desired cycle service level (%)	Z - score
<50	93	1.476
<=85	90	1.282
>85	87	1.126

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5.3 Input data correction:

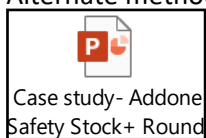
- Demand history is corrected to original demand region to remove IGST impact.
- Demand moved back to assigned DC based on customer master & based on distribution strategy (MRP type).
- Calculate standard deviation demand
- Lead time – Is the time taken by the process from its start until its conclusion. Here lead time means, the time taken from order placement by customer until its fulfilled.
- Transit time – Is the time taken by the parts to reach the destination during transit.
- CSL – Percentage of occasions on which order is delivered as promised and it is a matrix that measures the quality of customer service.

5.4 Fine-tuning

- Calculate for DC
- Seasonal inputs
- GTm inputs
- TCT targets

5.5 Next steps

1. Upload to PCD (Work Instruction)
2. Pick in Addone
3. Alternate method for safety stock (Add*one)



5.6 Review Mechanism

Topics	Frequency
Revision of Safety Stocks	Every 3 months
Safety stock correction and updation in system	Every 3 months
Safety stock review for Seasonal parts	Every 3 months

6. Attachments:

C4-AA-235 Stocking Policy
C4-APAC-049 Inventory Management

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IMPORTANT QUESTIONS and ANSWERS: -

1. What is Material Classification?

Answer:

Material Classification defines the assignment of materials to specific planning logic in Bosch IAM.

It determines:

- Applicable MRP type
- Stocking strategy
- Safety stock parameters

These parameters drive inventory and replenishment control.

2. What is the purpose of Material Classification?

Answer:

Material Classification ensures that each material follows the correct planning and replenishment logic.

It determines:

- How inventory is planned
- How stock is replenished
- How safety stock is maintained
- How materials flow through the distribution network

3. What is an MRP Type?

Answer:

MRP Type defines the planning logic for a material in SAP.

It controls:

- Forecast relevance
- Safety stock maintenance
- Replenishment triggers
- Inventory planning behavior

4. What are the standard MRP types used in Bosch IAM?

Answer:

Standard MRP Types are:

MRP Type Planning Logic

VV	Make-to-Stock, statistical forecast-based
SP	Make-to-Stock, manual forecast
OP	Make-to-Stock, fixed safety stock

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MRP Type Planning Logic

PD Make-to-Order

ND No planning

5. What is MRP Type VV?

Answer:

VV is a Make-to-Stock planning type based on statistical forecasting.

Characteristics:

- Used for fast-moving parts
- Stocked across all LDCs
- Safety stock maintained at all locations
- Customer orders fulfilled from the nearest warehouse

6. What is MRP Type SP?

Answer:

SP is a Make-to-Stock planning type based on manual forecasting.

Characteristics:

- Used for slow-moving parts
- Demand planning aligned with OFTM and SMS
- Stocked only at BAN and NCR
- Orders routed based on shortest distance

7. What is MRP Type OP?

Answer:

OP is a Make-to-Stock planning type with fixed safety stock.

Characteristics:

- No forecast is used
- Purchase Requisition is triggered when stock falls below safety stock
- Used for ADS, TCT and VOR scenarios

8. What is MRP Type PD?

Answer:

PD is a Make-to-Order planning type.

Characteristics:

- Order-based planning
- Follows globally defined ABC logic
- Inventory is planned against customer demand

9. What is MRP Type ND?

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Answer:

ND means No Planning.

It is used for:

- Obsolete materials
- Non-planning-relevant materials

10. Why are MRP Types important?

Answer:

MRP Types directly impact:

- Forecasting
- Replenishment
- Safety stock calculation
- Material flow within the distribution network

11. What is a Stocking Strategy?

Answer:

Stocking Strategy defines how inventory is positioned within the delivery network to achieve the targeted customer service level while balancing inventory and availability.

12. What is a Stocking Policy?

Answer:

Stocking Policy defines the stock level to be maintained at each delivery node based on the assigned MRP Type.

It supports:

- Inventory visibility
- Faster response to customer demand
- Optimized inventory cost

13. What is the difference between Stocking Policy and Stocking Strategy?

Answer:

Stocking Policy

Defines stock level to maintain

Based on MRP Type

Focuses on inventory quantity

Stocking Strategy

Defines inventory positioning

Based on service level requirements

Focuses on distribution network design

14. Who are Fast Movers?

Answer:

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Fast Movers are materials:

- Assigned MRP Type VV
- Stocked across all LDCs
- Supplied from the nearest LDC

The objective is high availability and shorter lead time.

15. Who are Slow Movers?

Answer:

Slow Movers are materials:

- Assigned MRP Types SP, OP, PD, or Imports
- Stocked only at designated locations (BAN and NCR)
- Routed according to distribution logic

16. Why are VV materials stocked across all LDCs?

Answer:

VV materials are stocked across all LDCs to:

- Improve availability
- Reduce customer lead time
- Fulfill orders from the nearest warehouse

17. Why are SP materials stocked only at BAN and NCR?

Answer:

SP materials are slow-moving parts.

To optimize inventory placement:

- Safety stock is maintained only at BAN and NCR
- Orders are routed to the nearest designated location

18. What is Safety Stock?

Answer:

Safety Stock is inventory maintained to buffer uncertainties in:

- Demand
- Delivery quantities
- Delivery dates

Its purpose is to achieve a defined service level.

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19. Why is Safety Stock maintained?

Answer:

Safety Stock is maintained to:

- Act as a replenishment trigger
- Manage demand fluctuations during replenishment lead time

20. What inputs are used for Safety Stock calculation?

Answer:

Inputs include:

- Stocking Policy
- Demand History (12 months)
- CSL Trend (12 months)
- Target CSL
- TCT
- Inventory
- Customer Lead Time
- Source
- Distribution Network
- SFA

21. What is the output of Safety Stock calculation?

Answer:

The output is:

Safety Stock at warehouse level for relevant MTS parts to ensure on-shelf availability against customer orders.

22. How many safety stock calculation methods are defined?

Answer:

The document defines four methods:

1. Demand variability without lead time fluctuation
2. Demand variability with lead time
3. Independent variation in demand and lead time
4. Dependent variation in demand and lead time

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23. Which safety stock method is used in Bosch IAM India?

Answer:

Method 2 is used for optimal safety stock calculation in IAM India.

Method 2:

Safety Stock = Z-score × Lead Time × Standard Deviation of Demand

24. What is a Z-score?

Answer:

Z-score describes the position of a value relative to the mean measured in standard deviation units. It is used during safety stock calculation to determine the service level target.

25. How are desired service levels linked to Z-scores?

Answer:

Current CSL Desired CSL Z-score

<50	93%	1.476
≤85	90%	1.282
>85	87%	1.126

26. Why is demand history corrected before safety stock calculation?

Answer:

Demand history is corrected to:

- Remove IGST impact
- Move demand back to the assigned DC
- Align demand with customer master and distribution strategy
- Calculate accurate demand variation

27. What factors are used during safety stock fine-tuning?

Answer:

Fine-tuning considers:

- DC calculation
- Seasonal inputs
- GtM inputs
- TCT targets

28. How often is safety stock reviewed?

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Answer:

Safety stock review activities are performed every 3 months.

This includes:

- Safety stock revision
- Safety stock correction and update
- Seasonal parts review

29. Why does Bosch IAM use different MRP Types?

Answer:

Different MRP Types are used because materials have different demand patterns, replenishment requirements, and stocking strategies.

This allows Bosch IAM to:

- Apply the correct planning logic
- Optimize inventory levels
- Improve availability
- Control replenishment efficiently

30. How does MRP Type influence inventory planning?

Answer:

MRP Type determines:

- Forecast usage
- Safety stock maintenance
- Replenishment triggers
- Material flow

Therefore, it directly influences inventory planning and stock availability.

31. Why are fast movers distributed differently from slow movers?

Answer:

Fast movers require high availability and quick delivery.

Therefore:

- VV materials are stocked across all LDCs

Slow movers have lower demand and are centralized to reduce inventory investment.

Therefore:

- SP, OP, PD and Imports are stocked only at designated locations

32. How does material classification affect customer service level?

Answer:

Material classification determines:

- MRP Type
- Stocking policy

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- Distribution strategy
- Safety stock level

These factors directly influence product availability and CSL performance.

33. What is the relationship between Safety Stock and CSL?

Answer:

Safety Stock is maintained to achieve the desired Customer Service Level (CSL).

Higher uncertainty in demand or lead time may require higher safety stock to maintain service levels.

34. Why is SFA used in safety stock calculation?

Answer:

SFA is one of the inputs used when determining optimum safety stock.

Forecast accuracy influences demand predictability and therefore affects the safety stock required to maintain service levels.

35. How do lead time changes impact safety stock?

Answer:

Lead time is one of the key inputs for safety stock calculation.

Longer lead times increase exposure to demand fluctuations and may require higher safety stock levels.

36. Why is demand history considered during safety stock calculation?

Answer:

Demand history is used to understand demand variability.

Demand variability is a key factor in determining the safety stock required to maintain service levels.

37. Why is the distribution network considered during safety stock calculation?

Answer:

Safety stock must support the actual material flow and stocking locations within the distribution network.

Therefore, distribution structure influences where inventory should be held.

38. What happens if safety stock is too low?

Answer:

Safety stock that is too low may reduce the ability to absorb demand fluctuations during replenishment lead time.

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Business impact:

- Reduced availability
- Potential service level deterioration

39. What happens if safety stock is too high?

Answer:

Excessive safety stock increases inventory levels beyond operational requirements.

Business impact:

- Higher inventory carrying cost
- Increased inventory investment

40. Why is safety stock reviewed every three months?

Answer:

Demand patterns, CSL performance, lead times, and business conditions change over time.

Quarterly reviews ensure safety stock remains aligned with current planning conditions.

41. How does stocking policy support inventory optimization?

Answer:

Stocking policy defines the inventory level maintained at each delivery node based on MRP Type.

This helps balance:

- Availability
- Inventory investment
- Customer service performance

42. What is the relationship between stocking policy and distribution strategy?

Answer:

Stocking policy determines how much inventory is maintained.

Distribution strategy determines where inventory is positioned.

Together they define inventory placement and material flow across the network.

43. Why are customer mapping exceptions allowed for SP materials?

Answer:

The document states that exceptions may be made for some customers based on distance.

The objective is to support efficient order fulfillment through the most appropriate stocking location.

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44. Why is customer distance important for SP order fulfillment?

Answer:

SP materials are stocked at BAN and NCR.

Customer orders are routed based on the shortest distance to the customer location to support efficient fulfillment.

45. How does safety stock support replenishment planning?

Answer:

Safety stock acts as a replenishment trigger and protects against demand fluctuations during lead time. This helps maintain inventory availability while replenishment is in progress.